

Hence, $F_1 = (2 \times 10^{-9})(2 \times 10^{-9})/4\pi\epsilon_0 d^2 = F/3$ [repulsive]

20.

Ans: A

$$F_{net} = F_e = qE$$

since E is constant and same charge $\rightarrow F_e$ is constant $\rightarrow$ acceleration constant

$$U_P = q\phi_P \text{ and } U_Q = q\phi_Q$$

The potential at Q is greater than P , since charge is negative, therefore $U_P > U_Q$.